

Tone-Tune Alignment in Mandarin Pop Music



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Research question

What we know

- Tonal languages rely on acoustic pitch to distinguish between words
- Singing also makes use of acoustic pitch

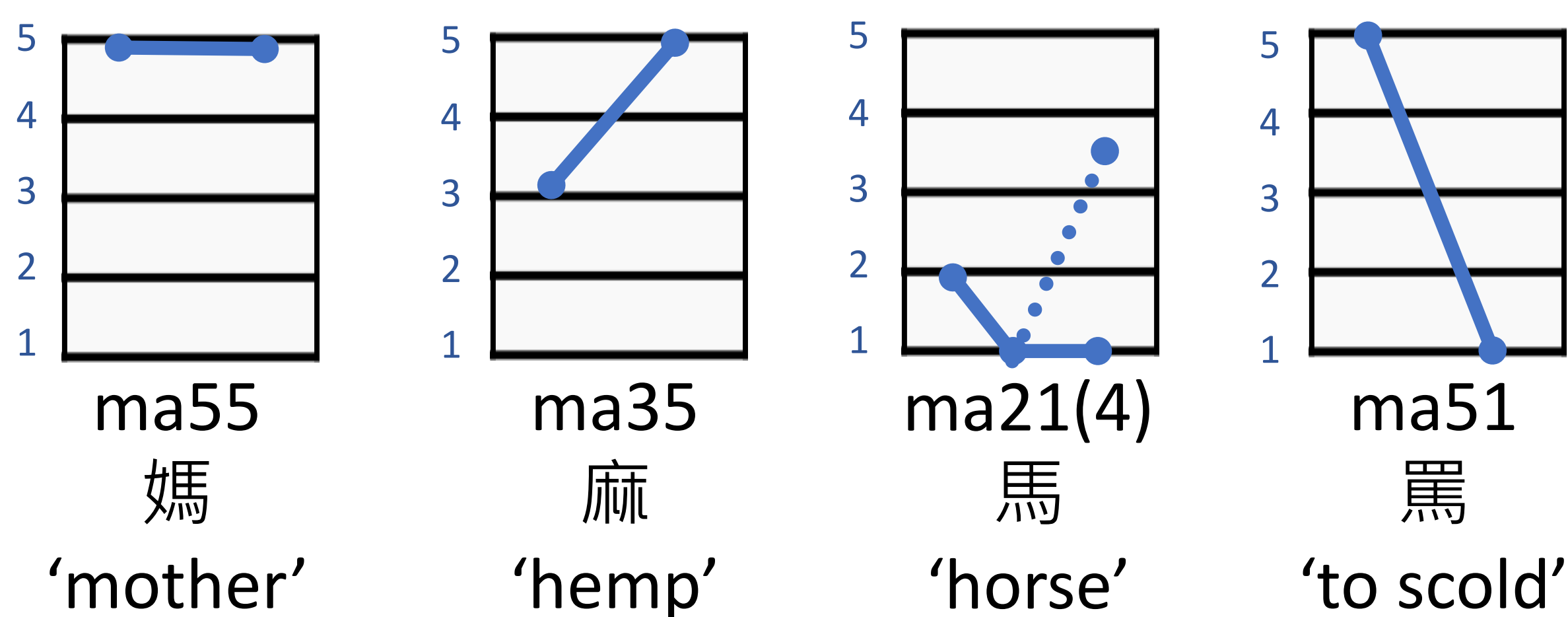
Question

- When singing, the linguistic tone is hidden, can listeners still identify the words?
- Is there tone-tune alignment in lyrics writing?

Previous work: Wee 2007, Kirby 2023

Mandarin tones

Chao tone numerals



Hypothesis

- There is some tone-tune alignment in Mandarin pop music at strong beats

Working assumptions:

1. Musical strong beat $\approx$ prosodic prominence
2. Strong beat note: local pitch maximum
3. Prominent syllable: perceptually salient, hence prioritized for tone-tune alignment

Prediction:

- At a musical strong beat,
If the melodic transition is ascending,
Then the tonal transition is also more likely to be a rise

Selected References: Kirby, James P. 2023. Comparative tonal text-setting in Mandarin and Cantonese popular song. Wee, L. H. 2007. Unraveling the relation between Mandarin tones and musical melody. Xu, Yi. 1999. F0 peak delay: When, where, and why it occurs.

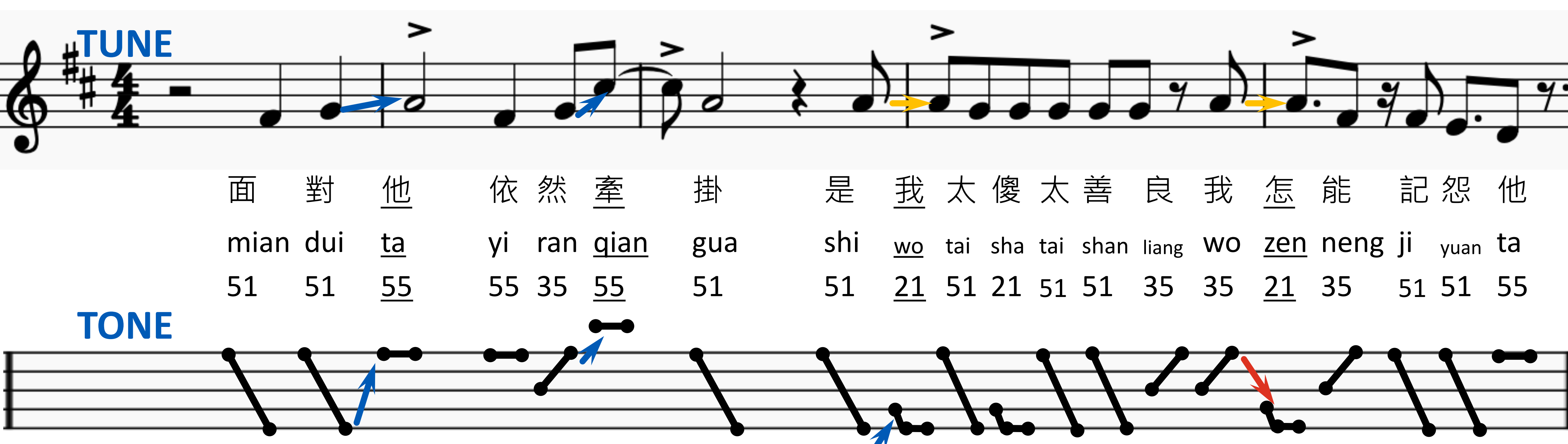
開門見山 Straight Up
Artist: 張惠妹 A-Mei

Composed by 阿弟仔 Adia
Lyrics by 林夕 Albert Leung



心酸的浪漫 Sad and Romantic
Artist: 那英 Na Ying

Composed by 張宇 Phil Chang
Lyrics by 那英 Na Ying



最浪漫的事 The Most Romantic Thing
Artist: 趙詠華 Cyndi Chao

Composed by 李正帆 Lee Cheng-Fan
Lyrics by 姚若龍 Daryl Yao

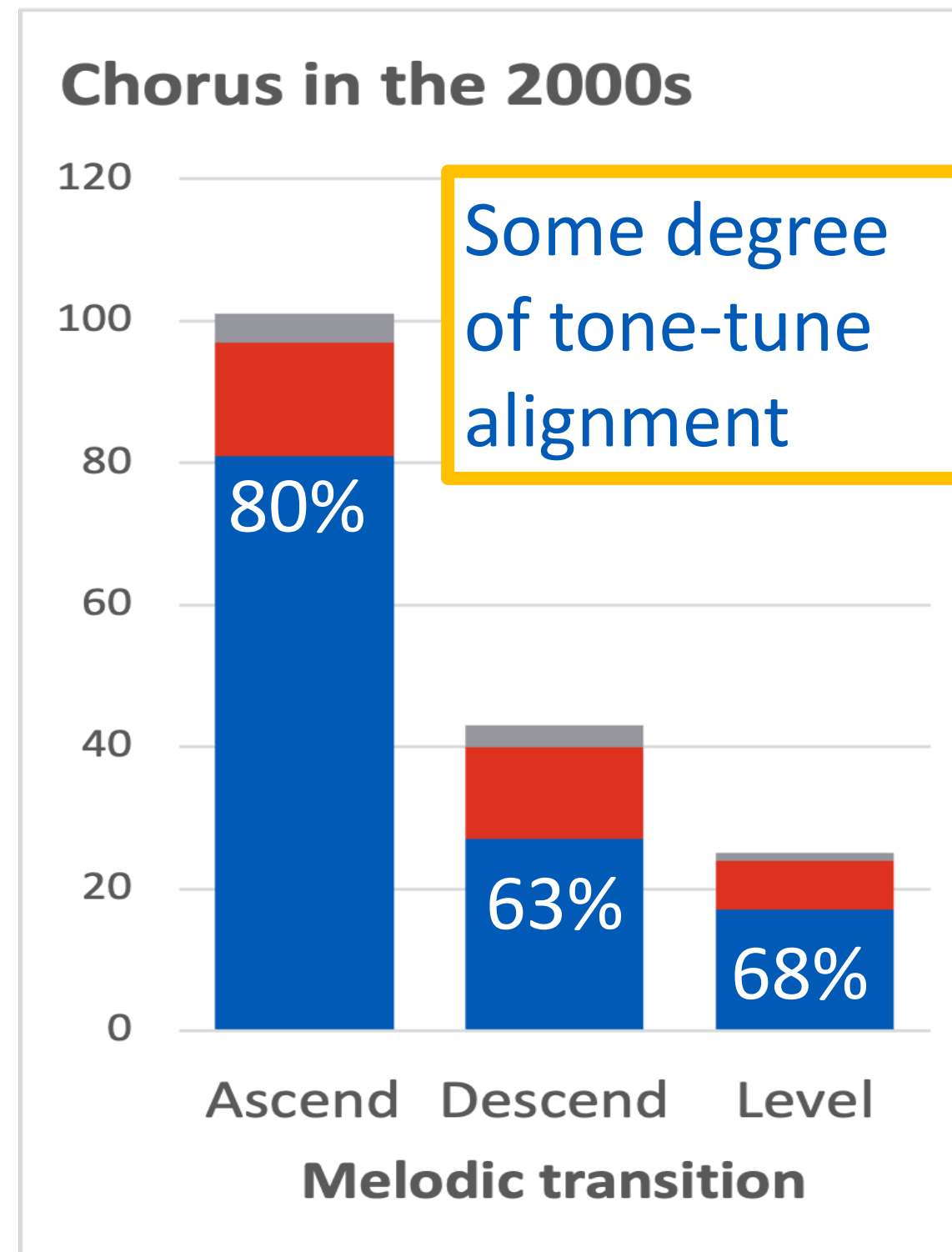
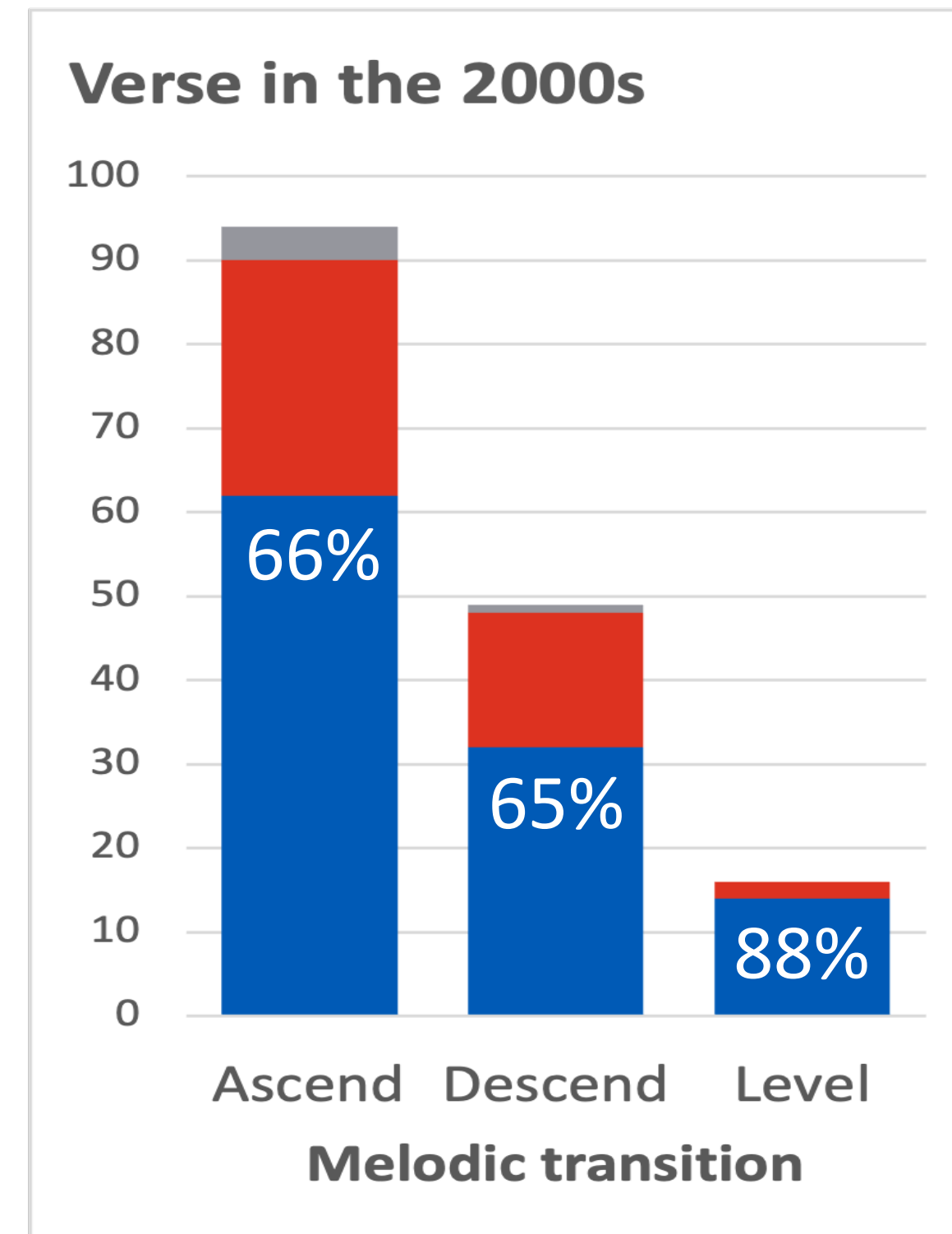
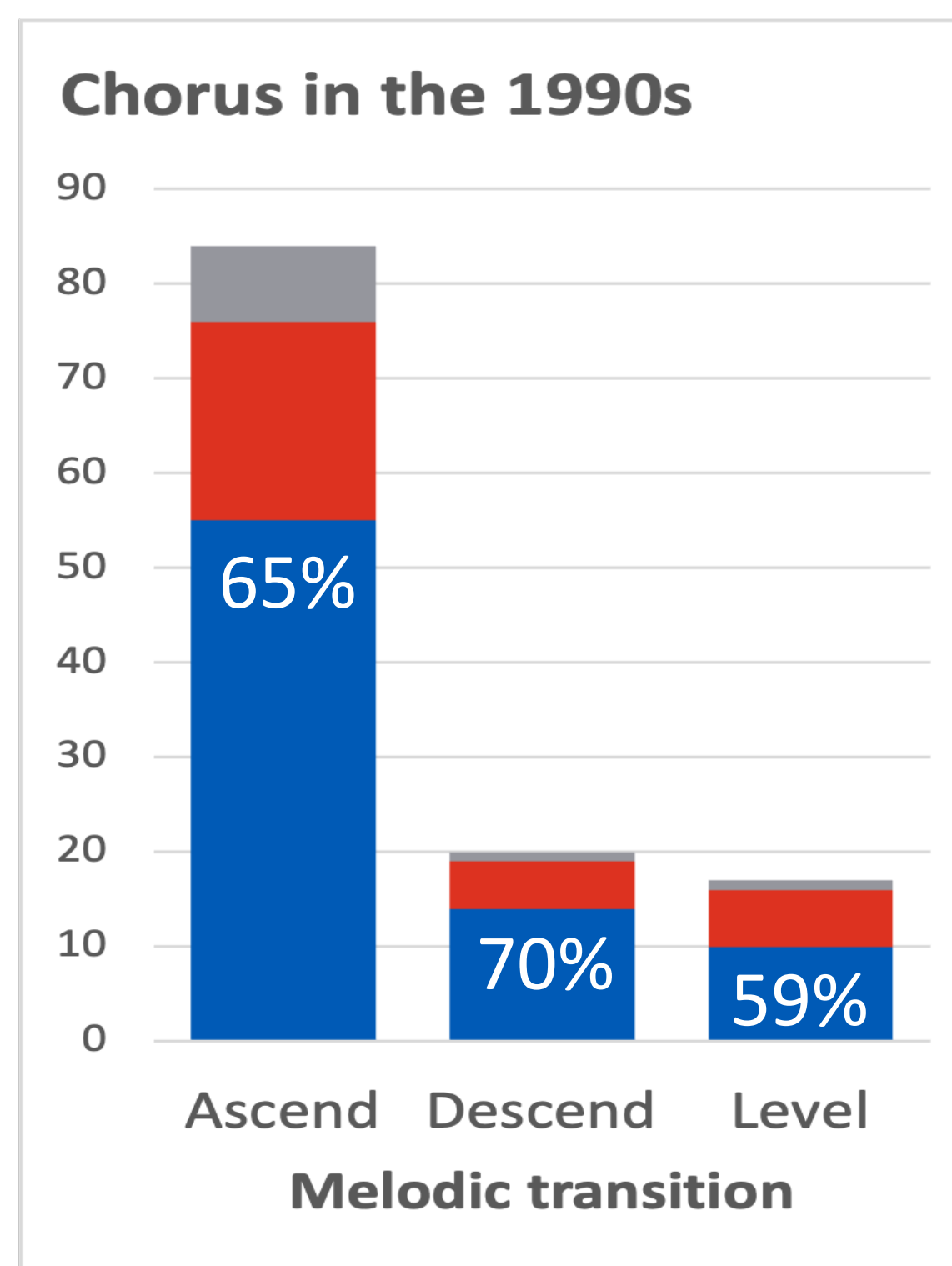
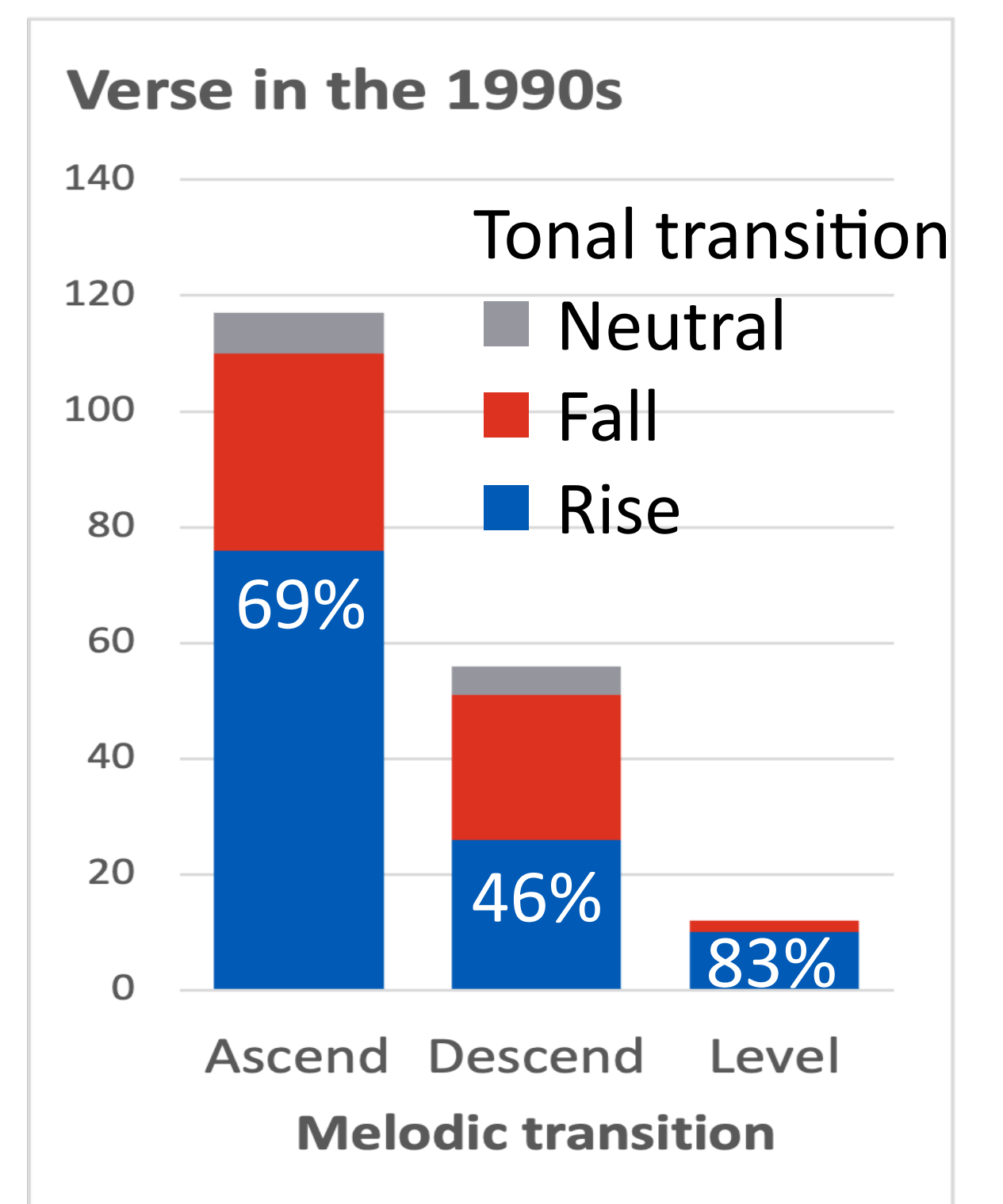


Method

Corpus study

- 16 Mandarin pop songs
- Won Golden Melody Awards for Best Lyrics
- 9 from the 1990s, 7 from the 2000s

Results



Confound

- Sample size too small
- Baseline bigram frequency of rising tonal transition: 62%

Future directions

- Grace notes added by singers
- Individual style of lyricists
- Tonal peak delay in Mandarin (Xu 1999)